Analytic Geometry: Lines, Circles, and Coordinate Tricks

N-20220911

§1.1 Line formulas

This note is a formula sheet and worked exercise collection for analytic geometry. The first part summarizes equations of a line.

If a line has inclination angle α, then its slope is

$$k = \tan\alpha, \qquad \alpha \neq \frac{\pi}{2}.$$

If it passes through two points

$$P_1 = (x_1, y_1), \qquad P_2 = (x_2, y_2),$$

then

$$k = \frac{y_2 - y_1}{x_2 - x_1}.$$

The point-slope form is

$$y - y_1 = k(x - x_1),$$

the slope-intercept form is

$$y = kx + b,$$

the two-point form is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1},$$

the intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1,$$

and the general form is

$$Ax + By + C = 0.$$

The note's diagram of intercepts reminds us that the intercept form is convenient only when the line meets both coordinate axes away from the origin.

§1.2 Parallel and perpendicular lines

For

$$l_1 : y = k_1 x + b_1, \qquad l_2 : y = k_2 x + b_2,$$

we have

$$l_1 \parallel l_2 \iff k_1 = k_2,\ b_1 \neq b_2,$$

and

$$l_1 \perp l_2 \iff k_1 k_2 = -1.$$

For lines in general form,

$$l_1 : A_1 x + B_1 y + C_1 = 0, \qquad l_2 : A_2 x + B_2 y + C_2 = 0,$$

parallelism is detected by proportional normal vectors:

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} \neq \frac{C_1}{C_2},$$

and perpendicularity is

$$A_1A_2 + B_1B_2 = 0.$$

This is just the dot product of the normal vectors.

§1.3 Direction and normal vectors

For a line

$$Ax + By + C = 0,$$

a normal vector is

$$n = (A, B),$$

and a direction vector is

$$d = (-B, A).$$

Thus the vector form through $P_0 = (x_0, y_0)$ is

$$\frac{x - x_0}{u} = \frac{y - y_0}{v}$$

if $d = (u, v)$, and

$$a(x - x_0) + b(y - y_0) = 0$$

if $n = (a, b)$.

This vector viewpoint is the easiest way to remember the formulas. The line is the set of points whose displacement from P_0 is perpendicular to a fixed normal vector.

§1.4 Angles, distances, and parallel lines

The angle between two nonvertical lines is given by

$$\tan\theta = \left|\frac{k_1 - k_2}{1 + k_1k_2}\right|.$$

Equivalently, using direction vectors d_1, d_2,

$$\cos\theta = \frac{|d_1 \cdot d_2|}{|d_1|\,|d_2|}.$$

The distance between points is

$$|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

The distance from a point $P(x_0, y_0)$ to a line

$$Ax + By + C = 0$$

is

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

For parallel lines

$$Ax + By + C_1 = 0, \qquad Ax + By + C_2 = 0,$$

the distance is

$$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}.$$

§1.5 Circle equations

A circle with center $C(a, b)$ and radius r has equation

$$(x-a)^2 + (y-b)^2 = r^2.$$

The general equation is

$$x^2 + y^2 + Dx + Ey + F = 0.$$

Completing the square gives

$$\left(x + \frac{D}{2}\right)^2 + \left(y + \frac{E}{2}\right)^2 = \frac{D^2 + E^2 - 4F}{4}.$$

Hence the center is

$$\left(-\frac{D}{2}, -\frac{E}{2}\right),$$

and it is a real circle exactly when

$$D^2 + E^2 - 4F > 0.$$

The handwritten note underlines this discriminant-like quantity because forgetting it leads to treating imaginary circles as real ones.

§1.6 A fractional-linear range problem

One example asks for the range of

$$\mu = \frac{x}{y-3}$$

under

$$|x| + |y| \le 1.$$

The idea is geometric. The equation

$$\mu = \frac{x}{y-3}$$

can be rewritten as

$$x = \mu(y-3).$$

For fixed μ, this is a line through the point $(0, 3)$. The range of μ is determined by which slopes make this line intersect the diamond

$$|x| + |y| \le 1.$$

Definition 1.6.1 (Pencil of lines)**.** A pencil of lines through a fixed point P is the family of all lines passing through P. Choosing a slope, or another one-dimensional parameter, gives a coordinate on most of this pencil.

Here the fixed point is $P = (0, 3)$. The parameter μ is not a random fraction; it is a coordinate on the pencil of lines through P. The range problem asks:

which lines in this pencil meet the diamond?

Because the diamond is convex and P lies outside it, the extreme values of μ occur at the two supporting lines from P to the diamond. These supporting lines touch the diamond at its upper-left and upper-right vertices:

$$(-1, 0), \qquad (1, 0).$$

Their parameters are

$$\mu = \frac{-1}{0-3} = \frac{1}{3}, \qquad \mu = \frac{1}{0-3} = -\frac{1}{3}.$$

Therefore

$$\mu \in \left[-\frac{1}{3}, \frac{1}{3}\right].$$

Remark 1.6.2 (Why this is not only a trick) — The algebraic fraction becomes transparent after dualizing the problem: points (x, y) in the diamond are replaced by lines through $P = (0, 3)$. The range is an interval because the diamond is convex, and the endpoints are supporting lines. This is a first glimpse of projective and convex dual thinking inside analytic geometry.

§1.7 A graph-slope counting problem

Another exercise asks for the number of x_i such that

$$\frac{f(x_i)}{x_i}$$

has a fixed value. Geometrically, the ratio $f(x)/x$ is the slope of the line from the origin to the point $(x, f(x))$. So the problem asks how many times a line through the origin can be tangent or secant to the drawn curve with a given slope. The answer is read from the graph, not from algebraic manipulation.

This is a useful change of viewpoint: ratios involving y/x often mean slopes of rays from the origin.

§1.8 A moving-line maximum

A later problem has moving lines through fixed points $A = (-3, 0)$ and $B = (1, 6)$, intersecting at P, and asks for the maximum of

$$|PA| \cdot |PB|.$$

The solution uses a right-angle configuration. Since the two moving lines are perpendicular, P lies on the circle with diameter AB. Then

$$|PA|^2 + |PB|^2 = |AB|^2.$$

Thus

$$|PA||PB| \leq \frac{|PA|^2 + |PB|^2}{2} = \frac{|AB|^2}{2}.$$

This is equality when $PA = PB$. With

$$|AB|^2 = (1+3)^2 + 6^2 = 52,$$

the maximum is

$$26.$$

§1.9 Triangle coordinate problem

The final worked example gives $A = (5, 1)$, the median line CM, and the perpendicular bisector BN, then asks for B and the equation of BC. The handwritten solution introduces unknown coordinates and uses two facts:

(i) a median means a midpoint condition;

(ii) a perpendicular bisector means equal distances or perpendicular slopes.

Coordinate geometry often works this way: translate each geometric phrase into an equation, then solve the resulting system.

§1.10 Coordinate tricks as changes of viewpoint

Analytic geometry often becomes simple after choosing better coordinates. A line is easiest when written in point-direction form,

$$p + tv,$$

while a circle is easiest when written by center and radius,

$$(x-a)^2 + (y-b)^2 = r^2.$$

Expanding too early hides the geometry. Completing the square recovers it.

§1.11 Power of a point

For a circle with center O and radius r, the power of a point P is

$$OP^2 - r^2.$$

If a line through P meets the circle at A, B, then

$$PA \cdot PB$$

is constant. This is a standard way to convert circle intersection geometry into algebraic length equations.

§1.12 The structural moral

Analytic geometry is not merely a bag of formulas. Each formula has a vector meaning: slope is direction, $Ax + By + C = 0$ encodes a normal vector, distance to a line is projection onto that normal vector, and circle equations are completed squares. The more one sees these meanings, the less one needs to memorize.

§1.13 Lines as affine equations

A line in the plane is an affine equation

$$ax + by + c = 0.$$

The vector (a, b) is normal to the line. Parallelism means proportional normal vectors; perpendicularity means the direction vector of one line is proportional to the normal vector of the other.

This turns many line problems into vector algebra.

§1.14 Circles and quadratic forms

A circle has equation

$$(x - a)^2 + (y - b)^2 = r^2.$$

Completing the square reveals the center and radius. More general second-degree equations are conics, and their classification depends on the quadratic part.

Thus circle problems are the simplest non-linear quadratic geometry.

§1.15 Fractional-linear transformations

A fractional-linear expression

$$y = \frac{ax + b}{cx + d}$$

is a Möbius transformation on the projective line when $ad - bc \neq 0$. Range problems for such expressions are often solved by inverting the equation and detecting excluded values.

§1.16 Analytic geometry through algebraic invariants

Analytic geometry translates geometric relations into algebraic invariants: normal vectors for lines, quadratic forms for circles/conics, and fractional-linear maps for rational range problems.

§1.17 Conics as quadratic forms

A conic in the plane has equation

$$ax^2 + 2bxy + cy^2 + dx + ey + f = 0.$$

The quadratic part is the symmetric matrix

$$Q = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

Orthogonal diagonalization of Q removes the xy term. Translation then attempts to remove linear terms. This is the linear-algebraic classification behind the elementary conic formulas.

The discriminant

$$b^2 - ac$$

distinguishes ellipse, parabola, and hyperbola types in the nondegenerate real case. In projective geometry, these distinctions are understood through rank and intersection with the line at infinity.

§1.18 Affine transformations

An affine transformation has the form

$$x \mapsto Ax + b, \qquad A \in GL(n).$$

It sends lines to lines and preserves ratios along a line, but not necessarily distances or angles. Many analytic-geometry simplifications are affine changes of coordinates.

Under affine transformations, ellipses may be transformed into circles, and general nondegenerate conics may be brought to standard forms. The elementary completion-of-squares method is one coordinate expression of this broader equivalence.

§1.19 Projective viewpoint on conics

Using homogeneous coordinates $[X : Y : Z]$, a conic is given by a homogeneous quadratic equation

$$\begin{pmatrix} X & Y & Z \end{pmatrix} A \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = 0.$$

Projective transformations act by

$$A \mapsto P^T AP.$$

This unifies ellipses, parabolas, and hyperbolas over the complex projective plane; the affine differences come from how the conic meets the line at infinity $Z = 0$.

§1.20 Duality

In projective geometry, points and lines have a dual relationship. A nondegenerate conic identifies points with polar lines. If a conic is represented by a symmetric matrix A, the polar line of a point p is

$$p^T Ax = 0.$$

This gives an advanced interpretation of tangents and chord relations.

§1.21 Returning to analytic geometry

The original formulas for lines, circles, and conics remain the computational core. The advanced layer says: lines are affine/projective objects, conics are quadratic forms, coordinate changes are group actions, and tangents/polars come from bilinear algebra.

§1.22 Distance formulas as dual norms

The distance from a point x_0 to a line

$$ax + by + c = 0$$

is

$$\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

The numerator evaluates an affine functional; the denominator is the norm of its linear part. In normed-space language, distance to a hyperplane uses the dual norm of the defining functional.

For a hyperplane $\varphi(x) = c$ in an inner product space,

$$\operatorname{dist}(x_0, H) = \frac{|\varphi(x_0) - c|}{\|\varphi\|_*}.$$

In Euclidean coordinates, $\|\varphi\|_*$ is the length of the normal vector.

§1.23 Conics and degenerations

A conic represented by a symmetric homogeneous matrix A may be nondegenerate or degenerate. Degenerate conics include pairs of lines, double lines, or points over the real affine plane. Algebraically, degeneracy corresponds to rank drop of A.

This gives a unified view of many analytic-geometry cases: a circle, ellipse, parabola, hyperbola, or pair of lines is controlled by rank, signature, and the line at infinity. Completing squares is a coordinate method for revealing those invariants.